

PQ Joysticks with the J1939 Message Protocol Controller Area Network (CAN) electrical output option use the J1939 Basic Joystick Message and, if required, the J1939 Extend Joystick Message to communicate device information.

Source Address:

The source address for J1939 joysticks is factory set according to the joystick position. The source address for J1939 Joysticks are as follows:

Joystick Location	Decimal Source Address	Hex Source Address
R = Right	51	33
L = Left	52	34
C = Center	53	35
A = Auxiliary	54	36

J1939 Basic Joystick Message -1

The J1939 Basic Joystick Message is used to transfer information about the measured status of the X and Y axis of a joystick base and the state of switches on the joystick grip.

Priority	Base PGN		PDU format		PDU Specific		Source Address		Data Field
	Dec	Hex	Dec	Hex	Dec	Hex	Dec	Hex	
3	64982	FDD6	253	FD	214	D6	*	*	8 bytes

CAN bus baud rate : 250KB

The resulting J1939 Basic Joystick Message PGN on the CAN bus is : 0x0CFDD6*

- * = Joystick hex source address

Data Field

The data field contains the joystick's output information. J1939 data fields contain 8 bytes of data.

Information in the data field

Byte#	1								2								3 and so on							
Bit#	1	2	3	4	5	6	7	8	1	2	3	4	5	6	7	8	1	2	3	4	5	6	7	8

Basic Joystick Message parameters and data field locations

Start Position (Byte/Bit)	Length (bits)	Parameter Name
1/1	2	Joystick X-Axis Neutral Position Status
1/3	2	Joystick X-Axis Lever Left Negative Position Status
1/5	2	Joystick X-Axis Lever Right Positive Position Status
1/7 -> 2/1-8	10	Joystick X-Axis position
3/1	2	Joystick Y-Axis Neutral Position Status
3/3	2	Joystick Y-Axis Lever Back Negative Position Status
3/5	2	Joystick Y-Axis Lever Forward Positive Position Status
3/7 -> 4/1-8	10	Joystick Y-Axis position

5/5	2	Joystick X-Axis Detent Position Status – N/A
5/7	2	Joystick Y-Axis Detent Position Status – N/A
6/1	2	Grip Button 4 Pressed Status
6/3	2	Grip Button 3 Pressed Status
6/5	2	Grip Button 2 Pressed Status
6/7	2	Grip Button 1 Pressed Status
7/1	2	Grip Button 8 Pressed Status
7/3	2	Grip Button 7 Pressed Status
7/5	2	Grip Button 6 Pressed Status
7/7	2	Grip Button 5 Pressed Status
8/1	2	Grip Button 12 Pressed Status
8/3	2	Grip Button 11 Pressed Status
8/5	2	Grip Button 10 Pressed Status
8/7	2	Grip Button 9 Pressed Status

Data Filed Examples

Byte	1							
Bit	8	7	6	5	4	3	2	1
	The 2 LSB of X-Axis Position		X-Axis Lever Right Positive Position Status		X-Axis Lever Left Negative Position Status		X-Axis Neutral Position Status	

Byte	2							
Bit	8	7	6	5	4	3	2	1
	MSB X-Axis Position							

Byte	3							
Bit	8	7	6	5	4	3	2	1
	The 2 LSB of Y-Axis Position		Y-Axis Lever Forward Positive Status		Y-Axis Lever Back Negative Position Status		Y-Axis Neutral Position Status	

Joystick X-Axis Neutral Position Status

00	Not On Neutral
01	On Netral
10	Error Indicator
11	Not Available.

Joystick X-Axis Left Negative Position Status

00	Not On Negative Side of Neutral
01	On Negative Side of Netral
10	Error Indicator
11	Not Available.

Joystick X-Axis Right Positive Position Status

00	Not On Positive Side of Neutral
01	On Positive Side of Netral
10	Error Indicator
11	Not Available.

*The full-scale output at the end of each linear zone will be 250 counts.

Joystick Button 1-7 Pressed Status

Bit Status	Remarks
00	Button Not Pressed
01	Button Pressed
10	Error Indicator
11	Not Available (No Button Installed)

J1939 Extended Joystick Message -1

The J1939 Basic Joystick Message is used to transfer information about the measured status of the X and Y axis of a joystick base and the state of switches on the joystick grip.

Priority	Base PGN		PDU format		PDU Specific		Source Address		Data Field
	Dec	Hex	Dec	Hex	Dec	Hex	Dec	Hex	
3	64983	FDD7	253	FD	215	D7	*	*	8 bytes

CAN bus baud rate : 250KB

The resulting J1939 Extended Joystick Message PGN on the CAN bus is : 0x0CFDD7*

- * = Joystick hex source address

Data Field

The data field contains the joystick's output information. J1939 data fields contain 8 bytes of data.

Information in the data field

Byte#	1								2								3 and so on							
Bit#	1	2	3	4	5	6	7	8	1	2	3	4	5	6	7	8	1	2	3	4	5	6	7	8

Basic Joystick Message parameters and data field locations

Start Position (Byte/Bit)	Length (bits)	Parameter Name
1/1	2	Grip X-Axis Neutral Position Status
1/3	2	Grip X-Axis Lever Left Negative Position Status
1/5	2	Grip X-Axis Lever Right Positive Position Status
1/7 -> 2/1-8	10	Grip X-Axis position
3/1	2	Grip Y-Axis Neutral Position Status
3/3	2	Grip Y-Axis Lever Back Negative Position Status
3/5	2	Grip Y-Axis Lever Forward Positive Position Status
3/7 -> 4/1-8	10	Grip Y-Axis position
5/1	2	Theta-Axis Neutral Position Status

5/3	2	Theta-Axis Counter Clockwise Negative Position Status
5/5	2	Theta-Axis Clockwise Positive Position Status
5/7 -> 6/1-8	10	Theta-Axis Position
7/3	2	Theta-Axis Detent Position Status – N/A
7/5	2	Grip Y-Axis Detent Position Status – N/A
7/7	2	Grip X-Axis Detent Position Status – N/A